

Master Game Design Document: Brewery Empire

1. Vision & Core Philosophy

- **Working Title:** Brewery Empire
- **Genre:** Macro-Management / Supply-Chain Strategy & Tycoon
- **Target Platform:** PC (Steam)
- **Core Loop:** Starting in the pre-industrial era, the player builds a global brewing empire by researching groundbreaking science, managing continental distribution networks, adapting to dynamic consumer tastes, and navigating historical world events.
- **Educational Intent:** Seamlessly integrate authentic brewing science (microbiology, organic chemistry, thermal dynamics) and industrial history into tooltips, event popups, research unlocks, and flavor text so the player implicitly learns how modern beer was created.

2. Visual & Art Direction

- **Perspective:** Macro-scale 2D/2.5D strategic management map (no micro-level interior building management or 3D pathfinding).
- **Map Style:** Textured paper/parchment strategy map featuring regional trade routes, resource nodes, and expanding city borders (*Victoria 3* inspired).
- **UI & Characters:** Hand-drawn illustrative character portraits for advisors, historical figures, and rival industrial magnates paired with period-accurate UI framing (*Anno 1800* inspired).

3. Gameplay Systems & Mechanics

Technology & Historical Science Tree

Progression spans four historical eras, with each node unlocking new mechanics while displaying historical "Science Snapshot" lore entries:

- **Era 1: Pre-Industrial Age (1750–1830)**
 - *Open-Top Fermentation:* Basic wooden vats. High wild yeast contamination risk.
 - *Malting Kilns:* Roasting barley for dark malts. *Lore: Unlocks dark ales, porters, and early stout styles.*
 - *Seasonal Harvesting:* Brewing restricted to cool winter months to avoid spoilage.
- **Era 2: Industrial Revolution (1830–1880)**
 - *Saccharometer & Hydrometer:* Measure sugar density and fermentation gravity.
 - *Patent Drum Roaster (Guinness Era):* Super-roasted unmalted barley. Enables light-bodied, highly dark stouts with long shelf-lives.

- *The Bohemian Pale Revolution (Pilsner Urquell, 1842)*: Soft water combined with pale malt and bottom-fermenting yeast. Creates golden lager, shifting world demand away from dark ales.
- *Ice House Logistics*: Storing winter lake ice in insulated cellars to enable summer lager brewing.
- **Era 3: Scientific Breakthroughs (1880–1920) — The Golden Age**
 - *Pure Yeast Isolation (Carlsberg, 1883)*: Isolating *Saccharomyces carlsbergensis*. Eliminates wild yeast contamination and raises base batch quality across the board.
 - *Mechanical Refrigeration (Spaten / Linde, 1873)*: Funding Linde's ammonia compressor eliminates seasonal brewing limits entirely.
 - *Pasteurization & Refrigerated Rail (Anheuser-Busch)*: Heat treating bottled beer and shipping via cold railcars opens continent-wide markets.
 - *Water Chemistry & pH Scale (Carlsberg Lab, 1909)*: Adjusting water hardness (soft vs. hard) and acidity to optimize mash efficiency.
- **Era 4: Modern Age (1920–Present)**
 - *Stainless Steel Vessels*: Replaces wood/copper to achieve maximum batch hygiene and consistency.
 - *Automated Bottling & Canning Lines*: High-throughput packaging for global logistics networks.

Custom Beer Recipe Builder & Dynamic Trends

- **Interactive Recipe Design**: Players tune parameters—Malt Ratios (Pale, Dark, Roasted), Hop Varieties & Bitterness (IBU), Color (SRM), Target Alcohol (ABV), Yeast Strains, and Water Profile (pH & hardness).
- **Scientific Feedback Loop**: Deviating from chemical thresholds (e.g., incorrect water pH for pale malt or insufficient sugars for target ABV) introduces off-flavors, shortening shelf life or lowering batch prestige.
- **Dynamic Consumer Market Trends**: City markets feature changing style preferences. Demands shift dynamically based on seasonal cycles (crisp lagers in summer, stouts in winter) and historical trend waves (the rise of golden lagers, temperance movements, or craft style resurgence).

Logistics Graph & Empire Management

- **Network Nodes & Edges**: Production Breweries $\rightarrow$ Central Warehouses $\rightarrow$ City Markets, connected via evolving transport edges (Horse Carts $\rightarrow$ Canals $\rightarrow$ Steam Railway $\rightarrow$ Cargo Ships).
- **Spoilage & Cold-Chain Management**: Distance, transit method, pasteurization status, and active refrigeration technology dictate how much inventory spoils in transit.

Victoria-Style Historical & Random Events

Pop-up events offer strategic choices, financial tradeoffs, and historical learning moments:

- **The Open-Source Pure Yeast Dilemma (1883)**: Share Emil Hansen's pure yeast

discovery freely for massive global prestige and market goodwill, or patent it for short-term monopoly profits.

- **The Great Ice Shortage:** A warm winter starves ice houses. Pay exorbitant transport rates for Scandinavian lake ice or halt lager production for the summer.
- **The Hop Blight:** Agricultural blight spikes raw hop prices. Substitute with herbal blends (risking prestige loss) or pay premium rates to maintain recipe integrity.
- **Temperance Movement Rises:** Local prohibition threats force choices between brewing low-alcohol session beers or spending heavily to lobby local authorities.

4. Technical Architecture (Unity / C# Framework)

- **Engine Choice:** Unity (C#) for its UI Toolkit layout flexibility, data serialization tools, and Steam SDK integration.
- **Decoupled Simulation Architecture:**
 - **Pure C# Core:** All business logic, graph routing, market calculations, event triggers, and tick updates are written in standard C# without dependencies on Unity visual components.
 - **Deterministic Tick Loop:** Game time advances via discrete steps (1 tick = 1 hour or day). Frame rate runs independently from simulation performance.
 - **Observer UI Pattern:** Unity scripts purely act as visual adapters. UI dashboards, parchment map nodes, and charts listen to simulation events and update passively when ticks complete.

5. Implementation Roadmap for AI / Developer Hand-off

1. **Phase 1: Pure Logic Foundation**
 - Build the standalone C# data structures for Nodes, Routes, Ingredients, and GameState.
 - Implement the discrete SimulationEngine and tick runner.
2. **Phase 2: Recipe & Market Systems**
 - Code the custom recipe parameter calculator (IBU, SRM, ABV, Quality rating).
 - Implement the market demand tick system and trend multipliers.
3. **Phase 3: Tech Tree & Event Engine**
 - Build the Directed Acyclic Graph (DAG) for research prerequisite nodes.
 - Implement the event engine with choice triggers and state modifiers.
4. **Phase 4: Unity UI Adapter**
 - Construct the parchment map visualization and UI Toolkit dashboards.
 - Bind event listeners to visual components.